

3.2 Defining the goal of your data visualisation ↕

When creating data visualisations, it is essential to know your data well, know your audience well and understand the differing chart types that are available. Because some charts are more appropriate for particular purposes, having an understanding of these factors will enable you to choose the best chart type for the task at hand. This will not only improve your analysis of the data but also your ability to communicate it effectively.

If you are going to use your chart to communicate with an audience, then the choice of chart that you use will depend on the message you want your audience take on board.

If the purpose of your chart is to explore data, then the type of chart that you should use will depend on the question(s) you want to find the answer to.

The type of data that you have may also influence your chart selection.

The key is to ask yourself is: what is your goal? Some common goals for charts are:

- *composition*. What comprises the whole of the entity that you are considering?
- *ranking*. What is the relative order of the items that you are considering?
- *correlation/relationship*. What is the relationship between two variables?
- *distribution*. How are the items dispersed?

As with most analytic tools, experimentation with different approaches is important.

□ Watch: Four question categories to help choose your chart types for data visualisation [6.12 mins]

This video explains how to choose your chart types for data visualisation.

This is useful as it explains different chart types that can be used for four different question types.

As you watch consider:

- Why does it matter what type of chart is used to visualise data?
- How can you use this understanding to improve your presentations?



Source: © Data Science and AI for Beginners and Kids (2019).

□ Activity: Check your understanding – choosing chart types

In the activity below, chart types are listed the left column. Drag the corresponding category (or categories) into the column on the right. Each category can be matched with multiple different chart types.

 **Attempt successfully submitted****Chart type****Category**

Scatterplot

Relationship - Distribution



Bubble chart

Relationship



Histogram

Distribution



Stacked column

Composition - Comparison



Pie chart

Composition



Stacked area

Composition - Comparison



Waterfall chart

Composition



Table

Comparison



Line chart

Comparison



Spider chart

Comparison



Column chart

Comparison

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